Social Media Advertisement Engagement Prediction

Group 10 Members: Andreana Argyres, Chiamin Yeh, Danyang Chen, Kaitlyn Hah, Dylan Sidhu

As technology continues to advance and more content becomes digitized, social media has become an increasingly important tool for companies to connect with and reach their target audiences. To maintain a competitive edge, it is essential for companies to continue evolving their digital marketing strategies to keep up in the ever-changing landscape of the Internet. Social media advertising has been recognized as an effective marketing strategy with up to three times higher engagement rates than traditional online ads[1]. It has many benefits that make it superior to traditional advertisements such as enhanced audience targeting, cost effectiveness, and real time engagement[1]. It is able to identify target audiences based on their age, gender, location, interests, and behaviors, increasing the likelihood of the user engaging with the product[1]. Additionally, it has a lower cost per impression than channels like TV and print advertising, while being more cost effective for companies by only charging per click[1].

In order for advertisers to better understand which ads are more likely to attract user attention, companies require insight into the key characteristics that lead to social media ad conversion and proper engagement. This can also provide insight into how different types of information contribute to online consumer behavior. Companies spend billions of dollars a year on advertising through various mediums in order to help launch their product into the market and the minds of the people. As social media is one of the more popular mediums of entertainment, understanding the conversion and the advertisement market within it is something that can be used to crucially improve many companies' marketing strategies.

There has been much research conducted about machine learning models in social media advertising, mainly in consumer behavior prediction. Recommendation systems are some of the most typical and well-known applications of machine learning in the field of marketing, and by analyzing user behavior patterns, the algorithms can provide personalized product recommendations to consumers which significantly improves customer version rates and satisfaction[2].

To assess the effectiveness of these models, Jin Lin investigated four machine learning models to measure their performance and demonstrate their practical applications in precision marketing[2]. Lin used the Online Shoppers Purchasing Intention Dataset (Version:2016) from the UCI machine learning library that included a variety of user engagement metrics such as page visits, stay times, bounce rates, exit rates, page value, and more. He used support vector machines (SVMs), extreme gradient boosting (XGBoost), categorical boosting (CatBoost), and backpropagation artificial neural network (BPANN) to predict consumers' purchase intention, evaluating their F1-scores, precision, accuracy, recall rate, and ROC AUC. Lin concluded that CatBoost with an accuracy rate of 93.4% and XGBoost with a precision of 93.5% were the best-performing models due to their ability to handle large–scale data and complex features in terms of prediction accuracy and generalization ability. While SVMs had the highest precision of 95.4%, it had a relatively low recall rate of 88.6%, making it effective in high-precision tasks but not as efficient at handling large data. BPANN also experienced limitations in large-scale data processing and was the worst performing model with a precision of 90.1%.

In this project, we compare and evaluate several binary classification methods to predict if a user will engage with a social media advertisement. Using different features about the advertisement, we want to learn from patterns between these variables and users' engagement and understand their correlations. Our results can provide insight into the key characteristics that lead to social media ad conversion and proper engagement. Additionally, our results can help companies understand what features are important to make an effective advertisement, while also allowing consumers to realize what influences their decisions.

Dataset Description and Processing

The Kaggle dataset utilized in this study comprises approximately 100,000 samples of social media advertisement interactions[3]. The primary objective is to predict the binary classification target variable, "Conversions," which represents if a user took an action aligning with the ad's goal, and identify the strongest predictors of audience engagement. The dataset's independent variables consist of 29 predictive features, comprising four distinct categories: demographic features (2): Age and Gender; engagement features (4): Impressions, Clicks, Sentiment Score, and Previous Interaction Score; categorical features (3): Location, Device Type, and Ad Category; and NLP (TF-IDF) Features (20): text-based advertisement content weights (tfidf_1 through tfidf_20), representing the vectorized textual elements of the ads.

After analyzing the continuous variables, we revealed that features such as Age, Impressions, Previous

Interaction Score, and Sentiment Score are almost perfectly uniformly distributed across the dataset. The only continuous feature exhibiting significant skewness is Clicks, which displays an exponential distribution. This indicates that while most advertisements receive a relatively low number of clicks, a select few generate an exceptionally high volume of user clicks. An examination of the categorical variables demonstrated that the demographic distributions are very well-balanced. However, our analysis highlighted a severe class imbalance within the target variable itself. 90% of the observations belong to the users who converted, whereas only a small fraction belongs to the users who did not convert. Furthermore, a correlation matrix was generated for the continuous features, revealing that the dataset's variables are largely uncorrelated. The singular exception is a positive correlation of 0.65 between Clicks and Impressions.

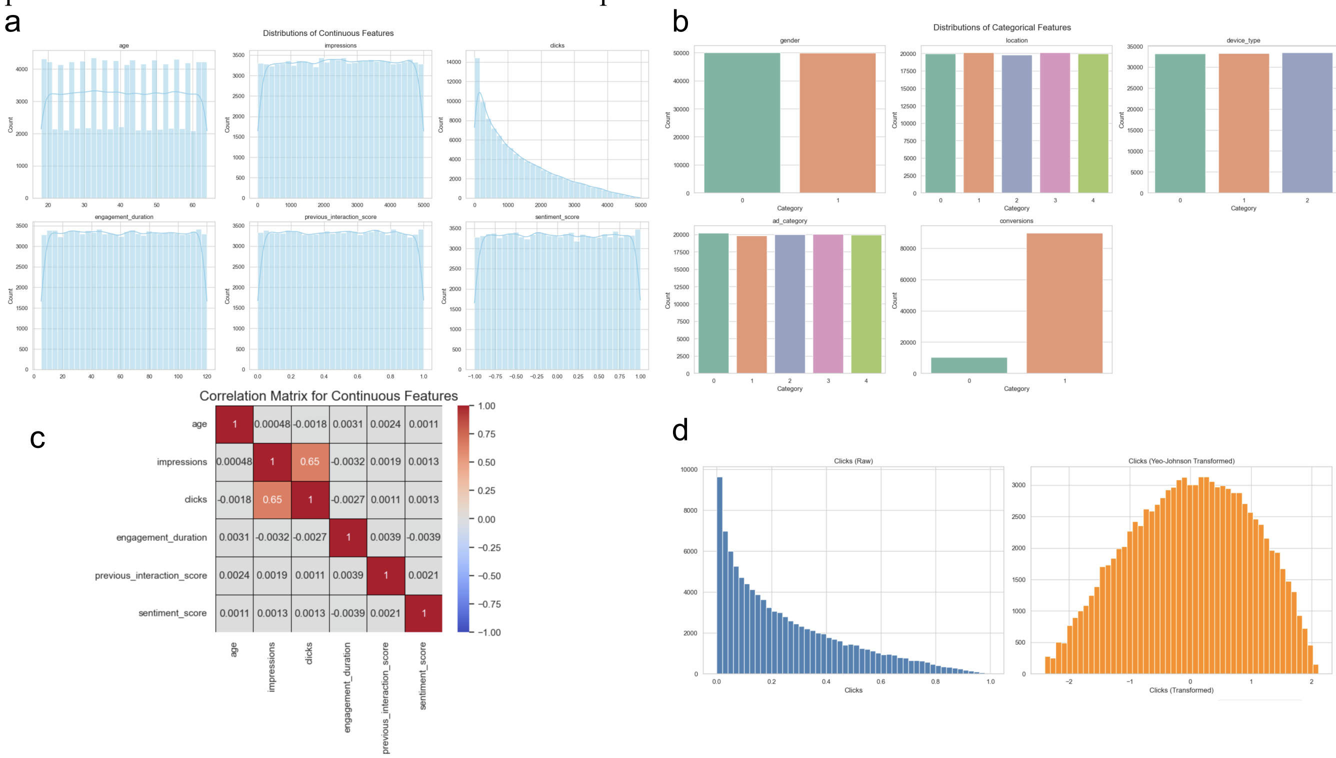

Figure 1: Exploratory Data Analysis. a, Histograms of the continuous variables. b, Bar plots for the frequency of values in the categorical features. c, Correlation matrix for the continuous features. d, Distribution of clicks before (shown in blue) and after (shown in orange) the Yeo-Johnson transformation.

To appropriately format the raw data for machine learning models, several critical preprocessing steps were implemented. First, non-predictive identifiers such as user_id, ad_id, and interaction timestamps were dropped from the dataset. To ensure that the categorical features did not introduce unintended bias into the models, one-hot encoding was applied to all categorical variables, excluding the target variable. Finally, to standardize the range of the independent variables, Min-Max Scaling was utilized to normalize the continuous features, ensuring they are all on the same scale. This allows neural networks to converge faster during training and is fundamental for distance-based algorithms. Additionally, due to the highly skewed distribution of the Clicks feature, a Yeo-Johnson transformation was utilized for one of the SVM models to normalize the Clicks distribution and improve model performance. The Yeo-Johnson transformation is a variation of the Box-Cox transformation that can handle zero values and was used since log transformation resulted in a left-skewed distribution. For all models, we used a ratio of 80:20 for the train-test split so that the models had enough data to learn from and be validated.

Proposed Solution and Experimental Results

Logistic Regression

We chose logistic regression because each coefficient tells us directly how a feature influences the probability of conversion. It is a natural baseline for binary classification and quick to train. Because the dataset is imbalanced (89.8% converted vs. 10.2% not), we applied `class_weight='balanced'` so the model does not just predict the majority class. For validation we used 5-fold stratified cross-validation, scoring on F1 rather than accuracy. We tuned regularization type (L1 vs. L2) and strength C over six values.

Penalty	C=0.001	C=0.01	C=0.1	C=1	C=10	C=100
L2	0.8068	0.8716	0.9216	0.9575	0.9772	0.9885
L1	0.8439	0.9401	0.9512	0.9522	0.9522	0.9522

Metric	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Score	98.24%	100.0%	98.04%	99.01%	1.0000

The model made zero false positives: every user it predicted as converted actually converted, and it caught 98% of all real conversions. Looking at Figure 2, clicks completely dominate with a value of 434, roughly ten times larger than Impression` at -44. All other features sit below 0.32. The negative sign on Impressions suggests ad fatigue: more views without a click slightly lowers conversion odds. L2 outperformed L1 (F1 0.989 vs. 0.952), meaning L1's sparse coefficients threw away useful signal. With C=100 performing best, overfitting was not an issue.

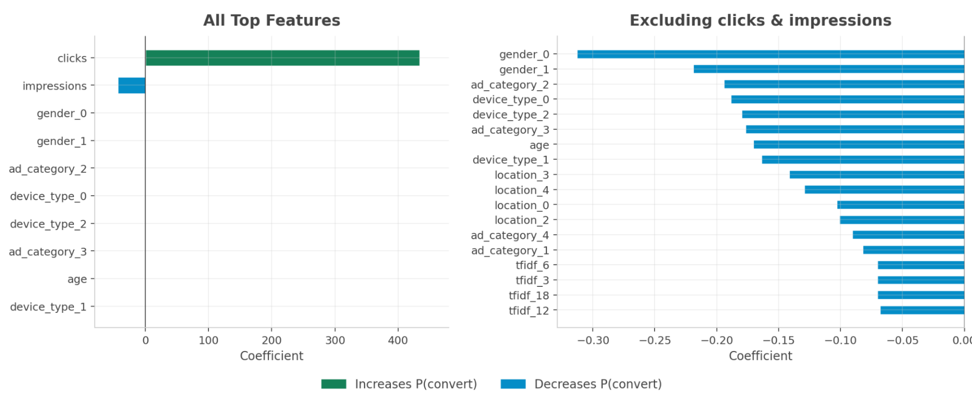

Figure 2. Feature Importance for the Logistic Regression Models.

Support Vector Machine (SVM) Classifiers

We next developed SVMs since they are useful for binary classification. We created both linear and non-linear models using the linear and RBF kernel because we did not know if our data was linearly separable or required more complex decision boundaries using the kernel trick. To tune the regularization parameter and kernel coefficient hyperparameters, we implemented random search with 5-fold cross validation, as random search is less computationally expensive than grid search.

Initially, our models performed well on the testing set despite the class imbalance, with an accuracy of 99.76% for the linear kernel and 97.88% for the RBF kernel. The models performed well on both classes but did the best on the majority class, the converted (1) class. For the not converted (0) class, the linear model had an F1-score of 0.9880, and the RBF model had an F1-score of 0.8922. For the 1 class, the linear model had a F1-score of 0.9880, and the RBF model had an F1-score of 0.8922. This indicates that the models had both high precision and recall. This is also seen in the confusion matrices, as only a small number of samples were misclassified for each class (Figure 3).

To address the class imbalance in our dataset to try to further improve performance, we utilized random oversampling and downsampling to equalize the number of samples per class for the training data. These methods can be advantageous since they do not make any assumptions about the data. Random oversampling randomly duplicates data for the minority class in the training set, while downsampling randomly samples a subset of the training data. To implement downsampling, we selected 8,145 samples from each class, the number of samples belonging to the minority class in the training set.

For the linear kernel, both random oversampling and downsampling boosted recall to 1.0000 for the 0 class but lowered precision to 0.8592 and 0.8567, respectively. Thus, both oversampling and downsampling improved the ability of the linear SVM to classify the 0 class, as all samples with the 0 class were classified correctly, but at the cost of more false positives (Figure 3). As a result, the overall accuracy of the linear model, 98.31% with random oversampling and 98.28% with downsampling, was slightly lower than the linear model with no class imbalance corrections.

For the RBF kernel, both random oversampling and downsampling significantly lowered performance, with an accuracy of 87.46% and 90.16%, respectively. Random oversampling drastically decreased the precision to 0.1642 and the recall to 0.0528 for the 0 class, and most of the samples with the 0 class were misclassified as the majority class (Figure 3).This occurred due to overfitting because when the model made predictions on the training data, the accuracy was 99.98%. This indicates that the model with the RBF kernel is only learning the duplicated examples of the minority class. Downsampling increased recall of the 0 class to 1.0000 but lowered the precision of the 0 class to 0.5119, as many samples with the majority class were misclassified (Figure 3). As a result, the reduction in examples of the 1 class hindered the ability of the RBF SVM to learn the patterns in the data needed to correctly classify the 1 class.

Since logistic regression showed that the number of clicks was an important feature, we tried making the distribution of clicks more normal with the Yeo-Johnson transformation to further enhance performance. Creating a more normal distribution can make it easier to separate out the data and fit the decision boundary. However, transforming the clicks distribution slightly lowered the performance for the linear SVM, which had an accuracy of 97.08% and did not have a significant impact on the RBF SVM's performance, which was 97.25%. The linear model with transformed clicks performed worse on the 0 class, with an F1-score of 0.8560, indicating that changing the clicks distribution made it harder for the model to linearly separate the data when predicting this class. The F1-scores for the RBF model were comparable with and without the Yeo-Johnson transformation, suggesting that making the distribution of clicks more normal did not help the RBF model separate the data.

Overall, our best SVM model used a linear kernel, a gamma value of 0.0001, a C value of 1000, and had no class imbalance corrections. This model had the highest accuracy and the smallest number of misclassifications for each class.

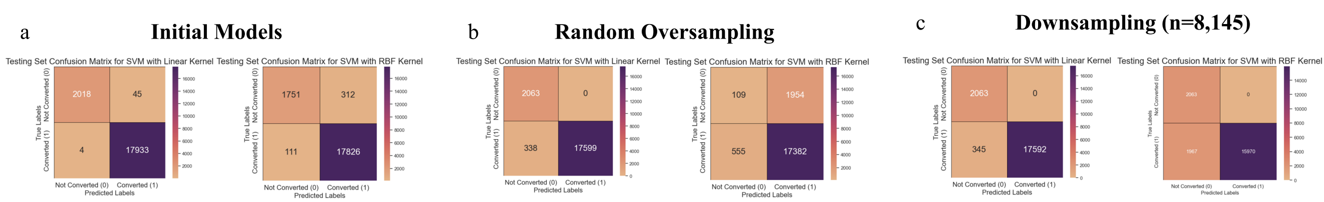

Figure 3. Confusion matrices for the SVMs created without class imbalance correction, with random oversampling, and with downsampling. Results are shown for both the linear and RBF kernels.

MLPs

For the final models, we developed MLP models because it can capture complex, non-linear relationships between predictors and the target variable. We decided to test models with 1 to 4 hidden layers as we thought that this would provide enough flexibility to capture the total complexity of the relationships in our data. We did hyperparameter tuning for the number of neurons per layer, model learning rate, epochs, and dropout rate, using 3 fold cross validation and random search to speed up processing. The complete search space is summarized in Table 1.

Layer #	Neuron Numbers (Used For All Layers)	Learning Rate	Epochs	Dropout Rate	Iterations (Random Search)
1	32, 64, 128	0.0001, 0.0005, 0.001, 0.005, 0.01	50, 100, 150	0, 0.1, 0.3, 0.5	40
2	32, 64, 128	0.0005, 0.001, 0.005	50, 100, 150	0, 0.1, 0.3, 0.5	50
3	8, 16, 32, 64, 128, 256	0.001, 0.01, 0.1	50, 100, 150, 200	0.0, 0.2, 0.3, 0.5	20

4	8, 16, 32, 64, 128, 256	0.001, 0.01, 0.1	50, 100, 150, 200	0.0, 0.2, 0.3, 0.5	20

Table 1: Hypertuning Parameters For Each Number of Layers of MLP

After fitting the model, it achieved an F1-score of 0.932 with an accuracy of 89%. However, the model returned 1 for all predictions with relatively high success due to the previously mentioned class imbalance. To address this, the weights of the classes were adjusted, and class 0 was rebalanced to weigh 4.86, about nine times as much as class 1's weight of 0.56. We chose to modify class weights over other balancing methods such as oversampling or undersampling because it allowed us to utilize all the data in the dataset without making up or duplicating data. The model was refit, achieving 91% accuracy and selecting 64 neurons, 0.005 learning rate, 100 epochs, and 0 dropout rate to be the best parameters with a CV F1-score of 0.952, the mean F1-score across all cross validation folds. The model was significantly worse at predicting class 0 than class 1, with a precision, recall, and F1-score of 0.65, 0.23, and 0.34 respectively for class 0, and 0.92, 0.99, and 0.95 respectively for class 1. We also conducted decision threshold tuning, testing thresholds from 0.1 to 0.9 at intervals of 0.01, resulting in 0.48 as the best threshold with a final test F1-score of 0.953 and the same accuracy of 91%.

For two hidden layers, it achieved an accuracy of 91%, selecting 128 neurons for both layers, 0.005 learning rate, 150 epochs, and 0.3 dropout rate as its best parameters, with a CV F1-score of 0.937. The optimal threshold was 0.62, resulting in a final accuracy of 95% and test F1-score of 0.972, a significant increase in performance across all metrics compared to one hidden layer as illustrated in Figure 4 and Figure 5.

Class	Precision	Recall	F1-Score
0	0.76	0.73	0.74
1	0.97	0.97	0.97

Figure 4: Classification Report for Best MLP 2 Hidden Layers at Threshold 0.62

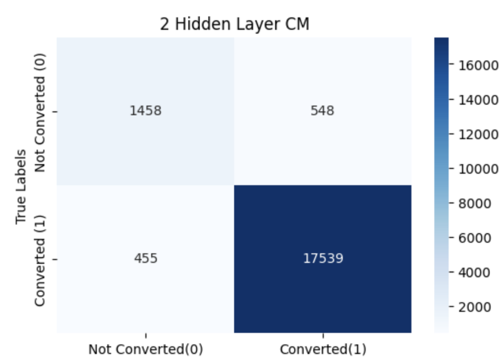

Figure 5: Confusion Matrix for Best MLP 2 Hidden Layers at Threshold 0.62

The model with three hidden layers originally achieved an accuracy of 74%. After fine tuning, the model had number of neurons being (64, 32, 26), learning rate of 0.001, dropout rate of 0.1 and 200 epochs. With changing the threshold from 0.5 to 0.45, the final accuracy for this model was 88%. It also had an F1-score of 0.458.

The four hidden layer model had an original accuracy of 10%, which is worse than the original accuracies of the other three models. The optimal parameters were selected as having the neurons (64, 32, 16, 8), learning rate of 0.001, dropout rate of 0.1, and 200 epochs. After fine tuning and changing the threshold from 0.5 to 0.9, the final accuracy was 10%,, with an F1-score of 0.1859.

As we can see, while performance increased from the 1-layer model to the 2-layer model, additional layers beyond that only decreased the accuracy and the overall performance of the model. One explanation is that additional layers add more complexity to the models, and while it can help better capture nonlinear relationships, too many layers adds a level of complexity that can become counter-intuitive to learning the data and may lead to being heavily class-biased. One of the limitations with these models is being able to use a very large range of parameters; this was limited by our computing ability and would be something that we would work on given more time with the modeling of MLPs.

Conclusion and Discussion

In this project, we investigated whether machine learning models can predict user engagement with social media advertisements using demographic, behavioral, categorical, and text-based features. Since the target variable, conversion, was highly imbalanced, we applied preprocessing steps such as removing non-predictive identifiers, one-hot encoding categorical variables, scaling continuous features, and testing imbalance-handling methods to improve model reliability. The dataset contained around 100,000 advertisement interactions and 29 predictive features, making it suitable for comparing different binary classification approaches.

Overall, logistic regression and the linear SVM performed the best. Logistic regression achieved strong predictive performance while also providing clear interpretability through feature coefficients. The linear SVM also performed very well, especially after downsampling, showing that simpler linear models were sufficient for this dataset. In contrast, MLP models performed the worst, and increasing the number of hidden layers generally decreased performance, likely because the additional complexity caused overfitting or stronger class bias rather than improving learning.

Feature importance analysis showed that clicks were by far the strongest predictor of conversion, meaning that user clicking behavior was the clearest signal of advertisement engagement. Gender also contributed to the prediction, while many other features had much smaller effects. These results suggest that in this dataset, user engagement behavior was driven more by direct interaction signals than by complex nonlinear feature relationships.

A key lesson from this project is that data cleaning and class imbalance handling are essential, but they do not always improve every model. For example, oversampling and downsampling helped some cases but hurt others, especially the RBF SVM. Therefore, model performance should be evaluated carefully using multiple metrics instead of relying only on accuracy. In conclusion, machine learning can be an effective tool for predicting social media advertisement engagement, but simpler and more interpretable models may be more useful than complex neural networks when the strongest predictive patterns are already captured by a few important features.

References

1. Griffin, S. (2024, November 7). Is Social Media Advertising Still Effective for Businesses? Americaneagle.com. https://www.americaneagle.com/insights/blog/post/how-effective-is-social-media-advertising
2. Lin, Jin. "Application of Machine Learning in Predicting Consumer Behavior and Precision Marketing." *PLOS ONE*, vol. 20, no. 5, 2025, e0321854. DOI: 10.1371/journal.pone.0321854.
3. Ziya. (2025, April 11). *Social Media Ad Engagement Dataset*. Kaggle. https://www.kaggle.com/datasets/ziya07/social-media-ad-engagement-dataset

Project Roadmap

The primary objective of this project is to build an accurate binary classification system capable of predicting user engagement with social media advertisements using behavioral, demographic, and advertisement-related features. By comparing multiple machine learning approaches and deploying the best-performing model within a web-based application, this project aims to demonstrate how data-driven analysis can improve digital advertising effectiveness and better understand consumer engagement behavior.

5/22: Problem Description & Background Research

- Define the prediction problem
- Explain the importance of social media advertising and engagement analysis
- Research related machine learning methods used in digital marketing and advertisement prediction
- Review similar studies and classification approaches

5/23: Dataset Understanding & Exploratory Data Analysis (EDA)

- Analyze the Social Media Ad Engagement Dataset containing 100,000 observations and its features
- Visualize feature distributions and relationships between variables
- Examine the balance of the "conversions" target variable
- Apply one-hot encoding for categorical variables
- Normalize numerical data
- Clean and preprocess dataset
- Complete exploratory analysis and data visualizations
- Gain better understanding of important features and trends within the dataset

---

5/26: Developing Prediction Models

Tasks / Action Items:

- Train binary classification models: logistic regression, Multi-Layer Perceptron(MLP) models, Support Vector Machine (SVM) models with different kernels
- Tune model parameters and compare preliminary performance results
- Identify promising model approaches

5/28: Model Evaluation & TestingTasks / Action Items:

- Evaluate models using accuracy, precision, recall, and F1-score
- Compare model performance on test data
- Analyze strengths and weaknesses of each model
- Select the best-performing classification model

5/30: Final Presentation

Prepare final report summarizing problem statement, dataset information, methodology, results, and conclusions

- Create presentation slides and video

6/6: Project Completion

- Upload code to final Jupyter Notebook
- Describe the problem, exploratory data analysis, models, and conclusions in the report
- Finalize project report and code

GitHub Link

All code used for this project can be found at https://github.com/DanielDanyang/ECS171-Group10